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High-performance flexible all-solid-state microbatteries based on solid electrolyte of lithium boron oxynitride

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HIGHLIGHTS

- LiBON solid electrolyte is integrated to Li/LiBON/$LiCoO_2$ microbattery.
- LiBON-based microbatteries demonstrate high cycling performance and rate capability.
- Excellent mechanical integrity against severe bending and twisting is achieved.

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ABSTRACT

Rapidly growing interest and demand for wearable electronics require the development of flexible and lightweight all-solid-state batteries as power sources that guarantee high performance and safety with the absence of the risk of fire or explosion that can occur with traditional liquid electrolyte systems. Herein, we successfully fabricate new flexible all-solid-state microbatteries integrating a solid electrolyte film of lithium boron oxynitride (LiBON) on a flexible substrate using sophisticated thin-film fabrication technology. The new microbattery of Li/LiBON/$LiCoO_2$ exhibits excellent mechanical integrity even under severe bending and twisting test conditions, enabling the realization of flexible microbatteries. The microbatteries demonstrate superior electrochemical cycling stability relative to conventional batteries, delivering an outstanding capacity retention of 90% on the 1000th cycle. Furthermore, operation at various temperatures from −10 °C to +60 °C and fast charging within 3–6 min are achieved. With various types of flexible substrates, the microbatteries can provide diverse opportunities for flexible and wearable electronics.

1. Introduction

Highly emerging technology of wearable electronics such as IoT, RFID tags, smart cards, sensors and small robots requires essential features of their batteries such as ultra-thinness, flexibility and light weight. Driven by the demand for wearable technology, recently the need of flexible microbatteries as miniaturized power sources has been rapidly increasing. All-solid-state microbatteries with Li^+-ion conducting inorganic solid state electrolyte satisfy these demands and provide high safety with the absence of the risk of fire or explosion that can occur with liquid electrolyte currently used in common batteries of portable electronic devices. All-solid-state microbatteries can fit into any shaped device, making the wearables universal and even more diverse than currently available ones. The most frequently employed solid electrolyte in all-solid-state thin-film batteries is lithium phosphorous oxynitride (LiPON) [1–13]. The intrinsic Li^+-ion conducting property of LiPON arises from a disordered structure caused by the partial substitution of O atoms in Li_3PO_4 with N atoms and the reduced electrostatic character in the bonding nature of the N–P–O compared to P–O. Li^+ ions in LiPON transport through N–P–O framework with ionic bonding character, resulting in a 10^{-6} S/cm level ionic conductivity at room temperature. The limitations of LiPON are however the limited ionic conductivity and self-discharge phenomenon due to the variable valence of P atoms [5,6]. Most LiPON-containing thin-film batteries exhibit a capacity decline upon long-term

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cycling, which is associated with resistance at the interface of cathode–LiPON and the limited Li^+-ion diffusion in the cathode [1,3,7]. To resolve the interfacial resistance problem, we employed a rapid thermal annealing (RTA) technology for cathode film deposition [10,11,14], which provides strong interfacial contacts between substrate and cathode film and between cathode film and LiPON film.

We recently reported a new oxynitride-based solid electrolyte material of lithium boron oxynitride (LiBON) of $Li_xBO_yN_z$ [15]. Thin films of lithium boron oxide composite glasses or their nitrided derivatives as solid electrolytes have been long studied, most of which were Li-lean in composition [16–18]. After the overall performance evaluation of the LiBON-integrated microbatteries with various LiBON composition from Li-lean to Li-rich (higher than 1 mol of Li atom relative to B atom), the best composition was determined to be $Li_{3.09}BO_{2.53}N_{0.52}$. The mole number (3.09) of Li atoms in our LiBON exceeds 2.64 Li of conventional LiPON ($Li_{2.64}PO_{2.81}N_{0.33}$), and the valence of boron is fixed as stable as +3. These are significant merits in view points of ionic conductivity and chemical stability of LiBON material, respectively. The ion conductivity determined using a blocking electrodes cell of Pt/LiBON/Pt was 2.3×10^{-6} Scm^{-1} at room temperature as previously reported [15], which is higher than that of LiPON (1.2×10^{-6} Scm^{-1}) reported by our group [7–13]. Nonetheless, the successful fabrication of a new microbattery system, into which a new film component of LiBON solid electrolyte is integrated, has been a long and tough process. The fabrication of microbatteries on a flexible substrate would pave the way to widen their integration into flexible and wearable electronics. Powering the device while maintaining the flexibility of a microbattery is a challenge. Numerous attempts have been devoted to develop flexible batteries on flexible substrates such as carbon nanotubes [19], polymers [20] and metals [21] and metal-coated polymers [22] but such batteries consisted of all bulk materials. We demonstrated the flexibility of LiPON-based microbatteries fabricated on mica at a few tens of micrometer thickness; mica is the transparent sheet silicate mineral that possesses good mechanical properties [12,13,15]. In this study, we demonstrate the successful fabrication of new flexible all-solid-state microbatteries of $Li/Li_{3.09}BO_{2.53}N_{0.52}$(LiBON)/$LiCoO_2$ on a flexible mica substrate using thin-film fabrication technology. The LiBON-based microbatteries exhibit excellent mechanical endurance even under severe bending and twisting test conditions, enabling a realization of flexible microbatteries. The microbatteires deliver superior electrochemical cycling stability compared to conventional microbatteries. The LiBON is believed to be superior alternative to LiPON in all-solid-state microbatteries.

2. Experimental section

Microbatteries of Li/LiBON/$LiCoO_2$ were fabricated by layer-by-layer deposition of first 250 nm thick Pt film on a 50 μm thick mica substrate with DC magnetron sputtering, and sequentially followed by films of 1.7 μm thick $LiCoO_2$ by RF-DC hybrid magnetron sputtering in 10 mtorr of Ar gas and 1.5 μm thick LiBON by RF magnetron sputtering, and 1.0 μm Li by thermal evaporation. Finally, Ni anode current collector and air-protective exterior coating of multilayered encapsulation were added. The $LiCoO_2$ cathode film was annealed by RTA [10,11,14] at 600 °C before a LiBON electrolyte film. A LiBON solid electrolyte film was deposited by RF magnetron sputtering at 13.56 MHz under 4 mtorr of N_2 atmosphere and at 3.53 W/cm^2 of power using a sintered target of $3Li_2O \cdot B_2O_3$ (Ulvac Materials, Inc.) with a diameter of 4 inch. Protective multilayered encapsulation was formed by layer-by-layer coating of polymer and oxide film on the microbatteries; first, a polyacrylate film was coated by UV polymerization, followed by coating of aluminum oxide film *via* chemical vapor deposition and sputtering, and coating of polyurea film by thermal polymerization. Then, one more layer of each aluminum oxide and polyurea films were added. The active electrode area of the microbattery was approximately 3 cm^2 (1.8 cm × 1.7 cm) and the total thickness was 10 μm including the protective exterior surface encapsulation.

The flexibility and mechanical integrity of our microbattery were tested by comparing the discharge capacity before and after twisting and bending tests of a microbattery (Li/LiBON/$LiCoO_2$)-integrated smart card, whose testing procedures followed the standard testing conditions (ISO/IEC-10373-1) of credit cards at the size of 8.5 cm (X-axis) × 5.5 cm (Y-axis). A microbattery was charged to 4.2 V after the initial charge-discharge cycling and moved to the bending tester. Bending test was carried out using a bend tester (Instron) by bening first the front side of a microbattery-integrated smart card to the directions of X-axis for 250 cycles and Y-axis for 250 cycles at the load displacement of 15 mm and 10 mm, respectively, at the speed of 30 cycle/min. The same tests were continued by bending the back side of the card to the directions of X-axis for 250 cycles and Y-axis for 250 cycles. Thus, total 1000 cycles of bending tests were conducted on the smart card. Then, twisting test was carried out using a twist tester (Instron). After fixing the microbattery-integrated smart card to the grips of a tester, the smart card was twisted to the directions of +15° and −15° for 1000 times at the speed of 30 cycle/min. After such harsh twisting and bending tests, the microbattery was separated from the smart card and cycled between 3.0 and 4.2 V at the rate of 1C, and a change in the capacity was compared to the one before these tests.

The as-deposited LiBON film was amorphous [15], which was also assessed from the film morphology by field emission scanning electron microscopy (FESEM, Inspect F, FEI) at 5 kV. The film composition was determined as $Li_{3.09}BO_{2.53}N_{0.52}$ by inductively coupled plasma atomic emission spectroscopy (ICP-AES) and elastic recoil detection-time of flight (ERD-TOF). To check N-doping, XPS analysis was conducted using a focused monochromatic Al K-α source for excitation and a spherical section analyzer (Thermo, MultiLab 2000). The pass energy was 20 eV, yielding an overall energy resolution of 0.1 eV. Spectra were recorded under a pressure of 5.0×10^{-9} torr, and the irradiated spot size was 200 μm. The B 1*s* and N 1*s* spectra were fitted using XPS peak fit after correcting the baseline using the Shirley method. The fitted peaks contained 91% Gaussian and 9% Lorentzian components. The goodness-of-fit was determined by the value of χ^2 to be below 2. The binding energy was calibrated based on the C 1*s* level at 284.6–284.8 eV. Depth profiling was performed using Ar ion sputtering at 2 kV.

The electrochemical charge-discharge cycling performance of fabricated microbatteries of Li/LiBON/$LiCoO_2$ was tested between 3.0 and 4.2 V at the rate of 1C (83.7 μA/cm^2 or 251 μA) for 1000 cycles after two formation cycles at 1C, and at a variable C-rate from 1C to 30C (2.52 mA/cm^2 or 7.56 mA), and at a variable operation temperature from −10 to +60 °C at the rate of 1C in a constant temperature chamber (TH-KE-065, JEIO Tech), using a multichannel battery cycler (WBCS3000, Won-A Tech).

3. Results and discussion

New microbatteries of Li/LiBON/$LiCoO_2$ were fabricated by layer-by-layer RF magnetron sputtering from the bottom by first depositing Pt on a mica substrate, and sequentially followed by films of $LiCoO_2$ and LiBON, and Li film by thermal evaporation, Ni anode current collector and air-protective exterior coating of multilayered encapsulation, as illustrated in Fig. 1a. The cross sectional image in Fig. 1b displays the resulting microstructure of the fabricated microbattery. The $LiCoO_2$ nanorods of approximately

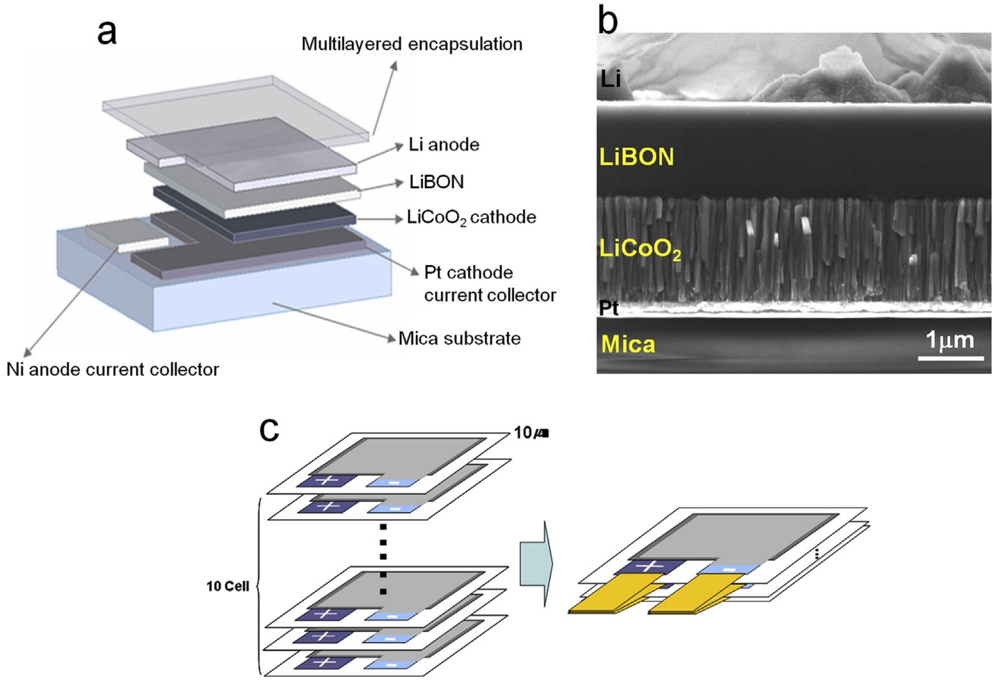

Fig. 1. Schematic for (a) the structure and (b) SEM image of cross-section view of the fabricated LiBON-based all-solid state microbattery, Li/LiBON/$LiCoO_2$, and (c) multi-stacks of the microbatteries for the required energy-density.

in ~100 nm in width that are normal to the mica substrate are oriented to the (101)/(104) planes as determined by Raman spectral analysis (Figure S1) [11–13]. The active electrode area of the microbattery was approximately 3 cm^2 (1.8 cm × 1.7 cm) and the total thickness was 10 μm including the protective exterior surface encapsulation. Fig. 1c illustrates the design and achievement of the target energy density. Multiple microbatteries are subjected to the array; a parallel arrangement and connection of ten of 0.5 mAh microbatteries enable a discharge capacity of 5 mAh. Similarly, a wide range of higher capacities can be achievable by fabricating the microbattery with a larger footprint and by integrating a few tens of those larger format microbatteries with multi-stacks.

A multilayered exterior encapsulation of microbatteries guarantees that they will be air-proof and water-proof, permitting an easy handling in air. Exterior encapsulation was achieved by multilayer coating with polymeric and oxide films *via* chemical vapor deposition and sputtering [13,23,24]. Fig. 2a demonstrates the transparency of multilayer encapsulation. Then, a yellow, shiny Li metal film, which is located right below exterior encapsulation, is visible. The flexibility of the microbattery is demonstrated as well in Fig. 2a. This flexible microbattery was embedded to a model smart card (Fig. 2b), whose thin plastic body is also flexible. Fig. 2c shows a futuristic model of an advanced smart card fabricated with an embedded light emitting diode (LED) and a piezoelectric switch for display and activation together with our microbattery as a miniaturized power source. When one physically taps the card with a finger, the piezoelectric switch is activated by vibration and the microbattery-powered LED is turned on. This demonstrates the realization of a microbattery-integrated smart card.

The flexibility and mechanical integrity of our microbattery were tested using a microbattery-integrated smart card by bending the card 1000 times and twisting it 1000 times, in which the microbattery is in a charged state after the initial charge-discharge cycle. Fig. 3 presents the detailed testing procedure following the standard testing conditions for credit cards (ISO/IEC-10373-1) and testing results. The capacity retention of the microbattery is observed to be higher than 99% of the initial discharge capacity even after such harsh twisting and bending tests. This demonstrates the excellent mechanical endurance and flexibility of our LiBON-based microbattery. This work is the first report of a flexible LiBON-based microbattery with significant flexibility. We believe that the fabrication of microbatteries on various types of flexible substrates will lead to the realization of diverse flexible and wearable electronics.

The composition and structure of the LiBON solid electrolyte film was characterized by spectroscopic analyses. The presence of N atoms in LiBON was monitored by X-ray photoelectron spectroscopy (XPS). Note that XPS data provide the chemical state of the top surface of the film. The curve fitting of B 1*s* (Fig. 4a) and N 1*s* (Fig. 4b) spectra was conducted, where the experimental peak shape was modeled by employing double-splitting patterns derived for B—N and O—B—N bonds, based on previously reported data for B_2O_3, BN and nitrided B_2O_3 [25–28]. The best fit to the experimental B 1*s* spectrum (Fig. 4a) showed that the top surface of the LiBON film possesses enriched B—N bonds at 190.6 eV with the presence of a peak attributed to O—B—N bonds at 192.6 eV. The presence of O—B—N bonds directly indicate successful N-doping in the position of O atoms in the lithium borate framework. The areal ratio (%) corresponding to the surface composition of O—B—N/B—N in the B 1*s* spectrum is 20/80. An additional clue for N-doping was assessed from the N 1*s* spectral curve fitting results (Fig. 4b). The peak attributed to B—N bonds is observed at 397.4 eV, and the presence of a peak at a relatively higher binding energy of 399.1 eV is ascribed to the O—B—N bonds. The areal ratio (%) of O—B—N/B—N peaks is 15/85, close to that of the B 1*s* spectrum. Fig. 4c exhibits the compositional profile as a function of depth (sputtered time) from the top surface to the bulk of the LiBON film. While the film surface is relatively N-rich as observed in the inset of Fig. 4b, the bulk is O-, Li- and B-rich.

The atomic concentration in the bulk of the LiBON film approaches the elemental analyses results obtained from ICP-AES. The structure of LiBON film, characterized using IR spectroscopy [15],

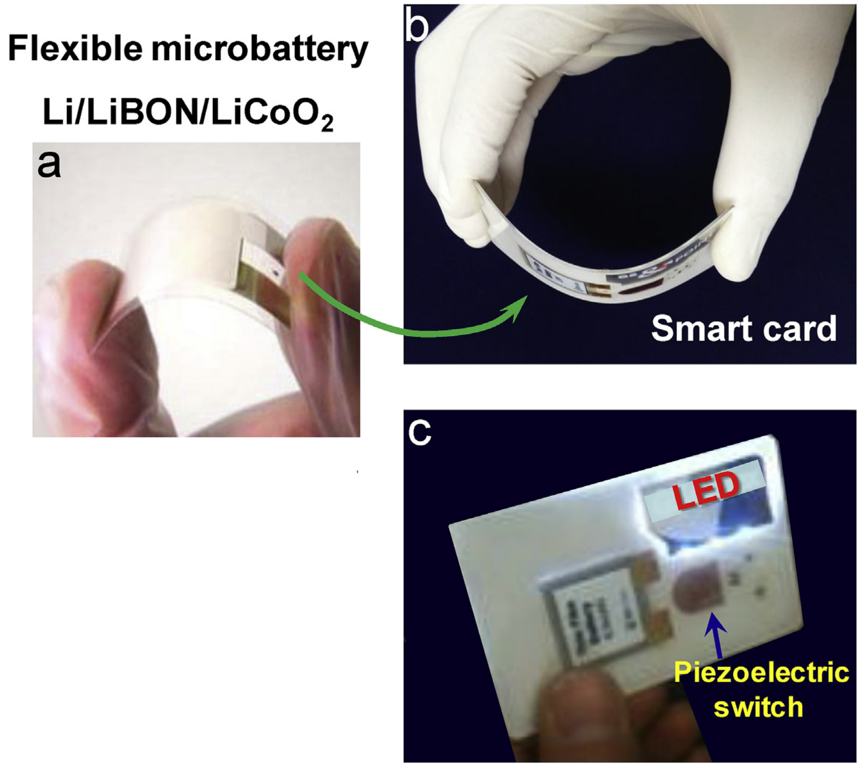

Fig. 2. Photos of (a) a flexible microbattery of Li/LiBON/$LiCoO_2$, (b) a microbattery-integrated flexible smart card, and (c) light-on of a LED of the smart card powered by the LiBON-based microbattery.

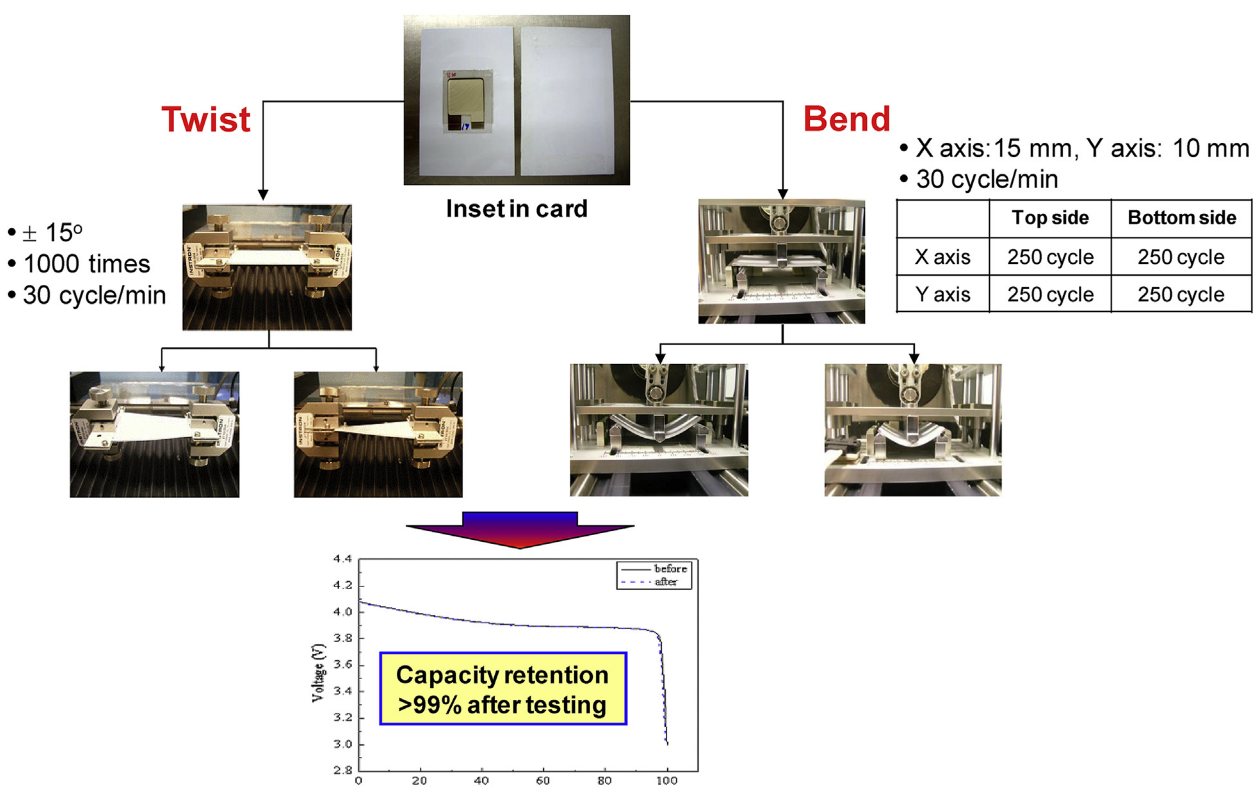

Fig. 3. Testing procedure of the flexibility of a smart card with LiBON-based microbattery, and comparison of capacity retention before and after twisting and bending tests.

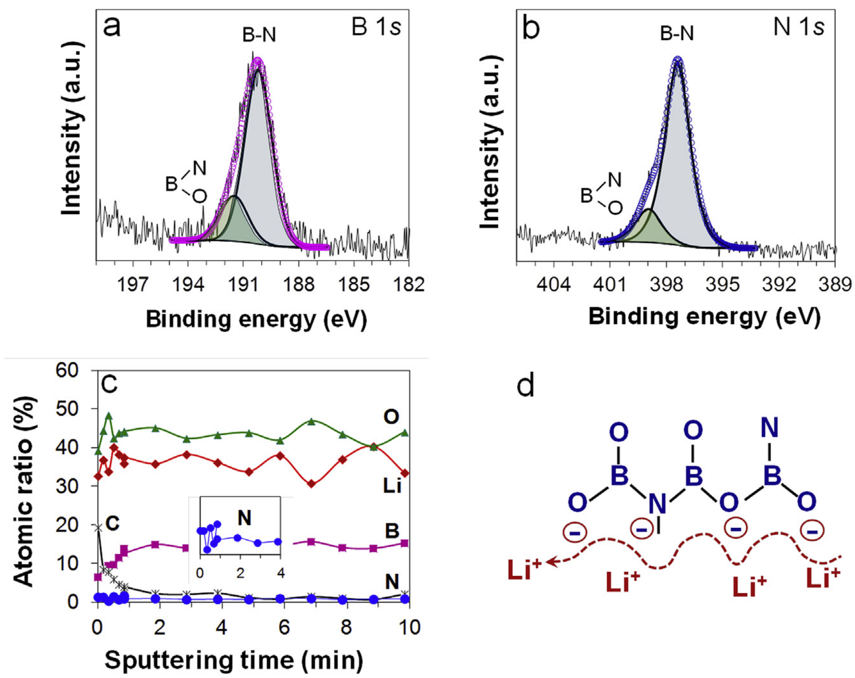

Fig. 4. XPS spectra of the (a) B 1*s* and (b) N 1*s* of LiBON solid electrolyte film and curve-fitting results, (c) XPS depth profiles of constituent atoms of LiBON film with the inset of the N atom, and (d) a schematic representation of the structure of LiBON solid electrolyte.

has BO_3 and BO_4 units and/or non-bridging oxygens probably from meta-, tri- and/or tetra-borate. The data dictate that N-doping and lithium (Li_2O) addition to B_2O_3 leads to the opening of the boroxol-ring of B_2O_3 followed by structural conversion to a disordered, glass-like structure. Fig. 4d represents the speculated molecular structure of LiBON, where Li^+ ions transport through the ionic bonds of -B-(O/N)$^-$···Li^+.

Electrochemical charge-discharge cycling tests of Li/LiBON/$LiCoO_2$ microbatteries were conducted in the voltage region of 3.0–4.2 V at the rate of 1C (83.7 μA/cm^2) at room temperature after two formation cycles. Fig. 5a shows a typical voltage profile of the first formation cycle of a microbattery; a high discharge capacity of 49.2 μAh/cm^2•μm and coulombic efficiency of close to ~100% are delivered, which is calculated based on the thickness (1.7 μm) of the $LiCoO_2$ film. Considering that specific gravimetric capacity of 135 mAh/g is generally obtained from $LiCoO_2$ bulk electrode in 3.0–4.2 V, the converted theoretical specific volumetric capacity in theory is 68.3 μAh/cm^2•μm, calculated using the density of $LiCoO_2$ (5.06 g/cm^3) and under the assumption that the film is perfectly dense. In reality, however, lower capacity than the theoretical value is obtained due to lower density of the sputtered $LiCoO_2$ film than the theoretical one, the possible presence of a very low fraction of spinel-like $LiCoO_2$ phase after RTA process, which yields lower capacity than layer structured $LiCoO_2$, and internal resistances. Differential capacity plots in Fig. 5b, obtained from the voltage profiles (Fig. 5a), exhibit highly resolved peaks. Prominent anodic peaks at ~3.93 V in the charge process is due to Li^+-deintercalation from the $LiCoO_2$ hexagonal structure, which is followed by a small anodic peak near 4.09 V by Li^+-deintercalation from the monoclinic phase and a peak near 4.2 V from the second hexagonal phase, consistent with earlier reports [11–13,29]. In the reverse process (discharge process), cathodic peaks at 4.19, 4.05 and 3.90 V are due to Li^+-intercalation to Li_xCoO_2. The very high structural resolution of Li^+-deintercalation and –intercalation, and the symmetric feature of the anodic and cathodic processes represent high crystallinity of the $LiCoO_2$ cathode film and reaction reversibility, respectively.

Fig. 5c demonstrates a very stable charge-discharge cycling performance of a Li/LiBON/$LiCoO_2$ microbattery at the rate of 1C. Discharge capacities of 49.2–44.2 μAh/cm^2•μm are obtained throughout the 1000 cycles. At the 1000th cycle, a significantly high capacity retention of 90% of the discharge capacity of the 1st cycle is achieved, which will guarantee a good cycle-life, in contrast to the lower capacity retention (82%) of our LiPON-based microbattery at the 1000th cycle even in a mild current density of 40 μA/cm^2 (~0.4C) [12,13]. The superior cycling stability of our LiBON-based microbattery compared to the LiPON-based one is ascribed to the use of the new solid electrolyte of LiBON.

Our microbattery performs well at room temperature (20 °C) and various operation temperatures from −10 °C to + 60 °C as displayed in Fig. 5d. At a low temperature of −10 °C, the discharge capacity at 1C decreases to 77% of the capacity (49.2 μAh/cm^2•μm) at room temperature, probably because of the limited ion conductivity of the LiBON solid electrolyte and sluggish and/or limited Li^+ diffusion into the $LiCoO_2$ cathode film at this low temperature. With an increase in the battery operating temperature, the discharge capacity at 1C gradually increases to 119% at +30 °C, and the capacity at +60 °C reaches 127% (62.5 μAh/cm^2•μm) relative to the capacity at room temperature. This behavior is likely related to the improved kinetics of Li^+ ion conduction and diffusion, and electronic conductivity of $LiCoO_2$ cathode film, and reduced interfacial resistances in the microbattery at elevated temperatures. The capacity at +60 °C is still lower than the theoretical specific volumetric capacity of 68.3 μAh/cm^2•μm, which was calculated under the assumption that the film is perfectly dense. The data emphasize the high thermal and chemical stability of LiBON solid electrolyte material and a high reliability in battery performance over a wide range of operating temperatures.

More importantly, the Li/LiBON/$LiCoO_2$ microbattery exhibits excellent rate capability, which has been recognized to be a hurdle for solid electrolyte-based rechargeable batteries. Fig. 5e exhibits the fast charging capability of the microbattery under a constant voltage of 4.2 V. Under these conditions, it takes just 5.9 min to

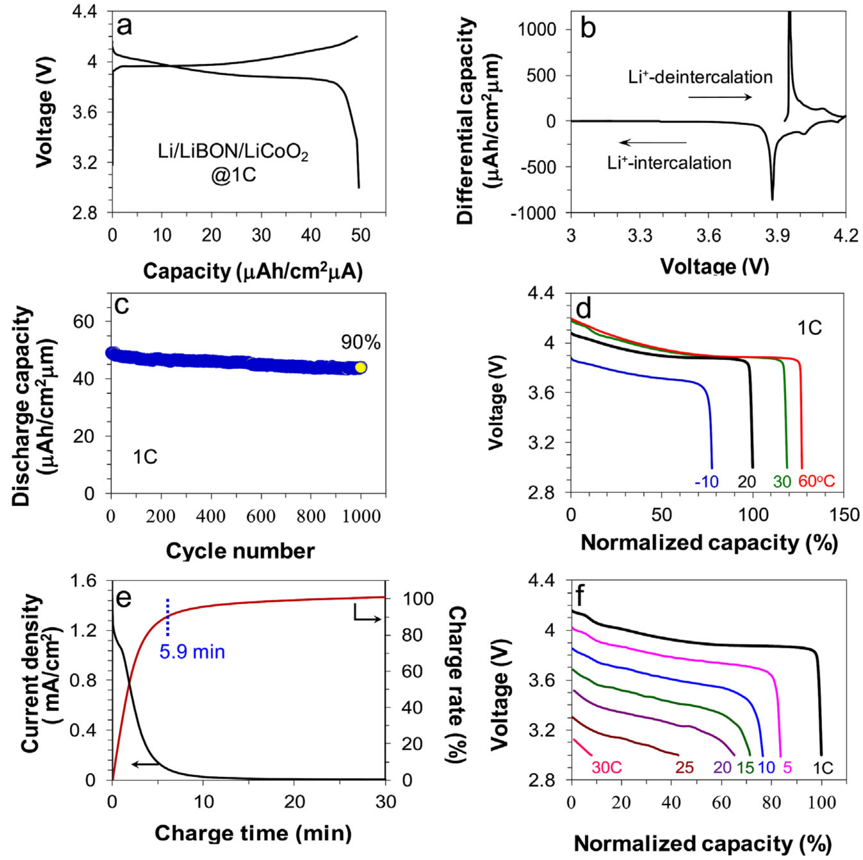

Fig. 5. Charge-discharge voltage profiles (a) and differential capacity plots (b) of the Li/LiBON/$LiCoO_2$ microbattery during the first formation cycle at 1C, and (c) cycling performance at 1C and 20 °C, and (d) the discharge profiles at 1C over a wide range of operating temperatures from −10 °C to +60 °C. (e) The attainment of 90% charge rate of the microbattery in 5.9 min and (f) high-rate performance from 1C to 30C at 20 °C.

achieve 90% of its capacity. This emphasizes the realization of a fast charging microbattery. Fig. 5f displays the high-rate performance of the Li/LiBON/$LiCoO_2$ microbattery, tested from the rate of 1C to 30C (2.52 mA/cm^2). The plateau of Li^+-intercalation is clearly visible up to the rate of 20C (i.e. charging in 3 min). The capacity delivered at the rate of 5C, 10C, 15C and 20C is 83%, 76%, 71% and 65%, respectively, relative to the capacity at 1C (49.2 μA h/cm^2•μm), indicating the excellent rate capability of LiBON-integrated microbatteries. This rate performance is comparable to ~68% capacity at 10C relative to the capacity at 0.73C of previous LiPON-based microbattery [13].

4. Conclusions

In this work, we report the successful fabrication of LiBON-based all-solid-state flexible microbatteries by sophisticated film fabrication technology. The use of a flexible substrate and strong interfacial adherence between constituent films provide mechanical integrity to the microbatteries for long repeated bending and twisting tests, which leads to excellent flexibility. The fabricated microbattery exhibits outstanding cycling stability, which is ascribed to the electrochemical stability of LiBON, and fast-charging capability. With various types of flexible substrates, the LiBON-based all-solid-state microbatteries hold great promise for diverse applications in a wide range of flexible and wearable devices that require a miniaturized power source with target energy and power densities.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.jpowsour.2016.07.114.

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